

Empirical Microstructure Alpha on Hyperliquid Perpetual Futures

Information-Coefficient Discovery, Cross-Symbol Validation, and
Hierarchical Signal Combination under Adverse Selection

NAT Quantitative Research Platform

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Abstract

We scan all 236 engineered features of the NAT platform against signed and absolute forward returns of Hyperliquid perpetual futures (BTC, ETH, SOL) at 100 ms tick resolution. Order-book imbalance is the dominant directional predictor, attaining a Spearman information coefficient (IC) of 0.40–0.47 at the 1–5 s horizon and decaying to near zero by five minutes (half-life ~ 15 –30 s). The same feature hierarchy holds across all three symbols, yielding eight approximately independent directional axes. The signal survives a battery of robustness checks — per-day, intraday, volatility-regime, bootstrap, and temporal out-of-sample — passing 7/8 for BTC and 6/8 for ETH and SOL. The single decisive negative result is that the *conditional* IC, measured on the subset of ticks where a directionally-correct mid-cross limit fill would occur, collapses from 0.45 to ~ 0.03 across all symbols. This is a structural adverse-selection barrier, not an estimation artifact, and it is the binding constraint of the project: signal quality is solved; profitable execution is not. We then present a three-layer hierarchical combiner (slow directional bias, gated fast entry timing, inverse-volatility sizing) that is the first architecture to address adverse selection directly rather than ignore it; it achieves walk-forward out-of-sample composite IC of 0.18 (BTC), 0.25 (ETH), and 0.36 (SOL) with cost-adjusted Sharpe > 0 on a thin two-day window, with appropriate caveats. Finally, a complementary candle-shape pattern-discovery engine (the “convolver,” using SVD + IC-gate + Benjamini–Hochberg FDR) is shown to be methodologically sound but data-starved: regime-robust discovery requires 6–24 months of data at the 60 s timescale and is effectively infeasible above the 15-minute timescale.

Contents

1	Introduction	1
2	Data and Methodology	2
3	Signal Discovery	2
3.1	Information-coefficient horizon profile	2
3.2	Full 236-feature cross-symbol scan	3
3.3	Directional/volatility orthogonality	4
3.4	Eight independent signal axes	4
3.5	Dead features	4

4	Signal Validation	4
4.1	Per-day, intraday, and volatility-regime stability	4
4.2	Bootstrap confidence and temporal out-of-sample	5
4.3	Conditional information coefficient: the adverse-selection barrier	5
4.4	Validation matrix	6
5	Hierarchical Signal Combination	6
5.1	Architecture	6
5.2	Walk-forward out-of-sample results	7
5.3	Comparison and honest assessment	7
6	Complementary Pattern Discovery: The Convolver	7
6.1	Current state and sample-size arithmetic	8
6.2	Why 50K candles is necessary but not sufficient	8
6.3	Viable niche and use as ML features	8
7	System Context	9
8	Discussion	9
9	Conclusion	9
	References	10

1 Introduction

NAT is a quantitative research platform for extracting alpha from Hyperliquid perpetual futures. A Rust ingestor computes 236 features per symbol every 100 ms; a Python stack performs discovery, validation, backtesting, and combination. For an extended period the platform’s ~ 25 trading algorithms were uniformly unprofitable. This paper synthesizes the sequence of internal studies that diagnosed the cause and reoriented the research programme.

The horizon mistake. Every algorithm in the system had been evaluated at a 100-minute forecast horizon using 5-minute bar aggregation. That aggregation destroys the predictive content of the microstructure features, which lives at the 1–5 second scale, while retaining only a weak residual ($IC < 0.07$). The features were never the problem; the evaluation horizon was. The reframing is simple but consequential: *the features work — the horizon was wrong*.

Contributions. This document fuses six internal reports into a single account:

1. **Tick-level signal discovery** (§3): an IC-by-horizon profile that localizes the directional signal to 1–5 s, and a full 236-feature cross-symbol scan that resolves eight independent directional axes and a clean directional/volatility orthogonality.
2. **Cross-symbol validation** (§4): eight robustness checks over 31 dates, culminating in the conditional-IC (adverse-selection) result.
3. **Hierarchical signal combination** (§5): a three-layer architecture that gates fast signals on a slow directional bias, with walk-forward out-of-sample results.

4. **Pattern-discovery data requirements** (§6): an analysis of why matched-filter candle-shape discovery needs years, not months, of data.

§7 situates the findings in the broader system, and §8 states the resulting research frontier.

2 Data and Methodology

Data. All studies use Hyperliquid perpetual futures at 100 ms tick resolution for three symbols (BTC, ETH, SOL). The primary discovery window is 2026-05-19 to 2026-05-21 (3 consecutive full days, $\approx 2.17\text{M}$ ticks per symbol). The validation study (§4) spans 31 dates from 2026-04-19 to 2026-06-04, of which 23 are valid (sufficiently complete) trading days.

Information coefficient. The IC is the Spearman rank correlation between a feature value at time t and the forward return over horizon h ,

$$r_{t,h} = \left(\frac{p_{t+h}}{p_t} - 1 \right) \times 10^4 \quad (\text{basis points}), \quad (1)$$

where p_t is the mid-price. *Directional* IC uses the signed return $r_{t,h}$; *volatility* IC uses the absolute return $|r_{t,h}|$. Spearman correlation is chosen for its non-parametric robustness to the heavy tails of tick returns. To suppress the strong autocorrelation of 100 ms sampling, evaluation points are subsampled every 100 ticks ($\sim 10\text{s}$), giving $\approx 21,700$ near-independent observations per symbol-horizon cell. Statistical significance is reported at $p < 0.01$ (two-sided); significant cells are marked with *. Horizons examined are $\{1\text{s}, 5\text{s}, 30\text{s}, 1\text{m}, 5\text{m}, 15\text{m}\}$.

3 Signal Discovery

3.1 Information-coefficient horizon profile

Table 1 reports directional IC versus horizon for the leading features on BTC. Order-book imbalance (level-1 and depth-weighted) is the strongest directional signal in the entire system, peaking at $\text{IC} \approx 0.45$ at 1–5s. Positive imbalance (more bids than asks) predicts upward movement within seconds; ask depth carries the mirror-image negative IC. The signal half-life is $\sim 30\text{s}$: by five minutes roughly 80% of the predictive content is gone.

Table 1: Directional IC (feature vs. signed forward return), BTC. * : $p < 0.01$.

Feature	1s	5s	30s	1m	5m	15m
<code>imbalance_qty_l1</code>	+0.447*	+0.453*	+0.257*	+0.184*	+0.089*	+0.056*
<code>imbalance_depth_weighted</code>	+0.427*	+0.434*	+0.255*	+0.185*	+0.081*	+0.037*
<code>raw_ask_depth_5</code>	−0.414*	−0.425*	−0.238*	−0.167*	−0.072*	−0.040*
<code>flow_aggressor_ratio_5s</code>	+0.112*	+0.112*	+0.074*	+0.051*	+0.009	−0.020
<code>raw_spread_bps</code>	−0.000	+0.006	+0.014	+0.023*	+0.073*	+0.139*
<code>regime_divergence_1h</code>	−0.020	−0.009	−0.035*	−0.043*	−0.058*	−0.074*

Two slow features behave oppositely, with IC that *grows* with horizon: `raw_spread_bps` ($\text{IC} = 0.14$ at 15m) and `regime_divergence_1h` (reaching $\text{IC} \approx 0.21$ at the 100-minute scale). These capture slow accumulation/distribution rather than microstructure, and explain the weak residual that the original 100-minute evaluation was inadvertently measuring.

3.2 Full 236-feature cross-symbol scan

Of 236 schema features, 207 are live (48 are uniformly NaN; see §3.5). Table 2 classifies the live features by behaviour for each symbol; Table 3 lists the top directional features cross-symbol at their 1–5 s peak. The imbalance cluster dominates uniformly across BTC, ETH, and SOL — the signal is universal, not symbol-specific.

Table 2: Feature classification counts by symbol.

Category	BTC	ETH	SOL	Description
<code>dir_fast</code>	29	29	28	Directional at 1–30s
<code>dir_medium</code>	11	11	10	Directional at 1–5min
<code>dir_slow</code>	7	3	14	Directional at 5–15min
<code>volatility</code>	68	75	68	Predicts magnitude, not direction
<code>weak</code>	11	9	7	Marginal signal
<code>no_signal</code>	33	32	32	Nothing at any horizon
<code>all_nan</code>	48	48	48	No data produced

Table 3: Top directional features (peak IC at 1–5s, all $p < 0.01$).

Feature	BTC	ETH	SOL	Horizon
<code>imbalance_qty_11</code>	+0.453	+0.447	+0.466	1–5s
<code>imbalance_qty_15</code>	+0.456	+0.431	+0.453	5s
<code>imbalance_notional_15</code>	+0.456	+0.431	+0.453	5s
<code>imbalance_orders_15</code>	+0.435	+0.438	+0.455	1–5s
<code>imbalance_depth_weighted</code>	+0.434	+0.410	+0.425	5s
<code>raw_ask_depth_5</code>	−0.424	−0.404	−0.415	5s
<code>raw_bid_depth_5</code>	+0.420	+0.405	+0.406	5s
<code>imbalance_pressure_ask</code>	−0.417	−0.387	−0.418	5s
<code>cross_obi_mean</code>	+0.354	+0.334	+0.331	1–5s
<code>micro_queue_position_bid</code>	+0.307	+0.282	+0.291	5s
<code>flow_vwap_deviation</code>	−0.287	−0.206	−0.188	1s
<code>micro_obi_velocity</code>	+0.192	+0.181	+0.208	1s
<code>ent_permutation_imbalance_16</code>	−0.170	−0.173	−0.177	1s
<code>flow_aggressor_ratio_5s</code>	+0.112	+0.109	+0.111	1–5s

Decay by symbol. Peak directional IC falls with horizon at a symbol-dependent rate: BTC $0.447 \rightarrow 0.456 \rightarrow 0.261 \rightarrow 0.187 \rightarrow 0.090 \rightarrow 0.056$; ETH $0.447 \rightarrow 0.438 \rightarrow 0.219 \rightarrow 0.160 \rightarrow 0.074 \rightarrow 0.031$; SOL $0.466 \rightarrow 0.412 \rightarrow 0.190 \rightarrow 0.135 \rightarrow 0.056 \rightarrow 0.036$ (at 1s/5s/30s/1m/5m/15m). The implied half-lives are ~ 30 s (BTC), ~ 20 s (ETH), and ~ 15 s (SOL): the thinner book discovers price faster.

3.3 Directional/volatility orthogonality

The separation between direction and magnitude is clean and consistent across all three symbols. Imbalance features have essentially *zero* volatility IC; the best volatility predictors — `hawkes_intensity` (IC ≈ 0.35 at 5s), realized volatility `vol_returns_5m/vol_returns_1m` (~ 0.32), `vol_parkinson_5m`, trade-intensity `flow_count_30s/flow_intensity`, and `toxic_effective_spread` — have essentially *zero* *directional* IC. Direction and magnitude are measured by disjoint feature sets.

3.4 Eight independent signal axes

Within the highly-correlated directional cluster, the report resolves eight approximately independent axes (Table 4). These are the inputs later combined in §5.

Table 4: Independent directional signal axes and representative features.

Axis	Representative feature	IC (1–5s)
Order-book imbalance	<code>imbalance_qty_11</code>	0.45
Raw depth asymmetry	<code>raw_bid_depth_5</code>	0.42
Cross-symbol imbalance	<code>cross_obi_mean</code>	0.35
Queue dynamics	<code>micro_queue_position_bid</code>	0.31
VWAP deviation (reversion)	<code>flow_vwap_deviation</code>	0.25
OBI velocity (momentum)	<code>micro_obi_velocity</code>	0.19
Imbalance entropy	<code>ent_permutation_imbalance_16</code>	0.17
Aggressor flow	<code>flow_aggressor_ratio_5s</code>	0.11

3.5 Dead features

The same 48 features produce no data on every symbol: whale-flow (12), liquidation-risk (13), concentration (15), and GMM-regime (8). These depend on a position/trade-attribution pipeline that was not yet wired during these studies; their repair is tracked separately as the feature-wiring work. Their absence does not affect the order-book signals, which are computed directly from the L2 book.

4 Signal Validation

The candidate signal — order-book imbalance at 1–5 s — is subjected to eight robustness checks. Seven are confirmatory; the eighth (conditional IC) is the decisive constraint.

4.1 Per-day, intraday, and volatility-regime stability

Per-day IC (Table 5) is stable across 23 valid dates. Intraday, the signal is present 24/7 with no dead zones (a slight dip at 12–16 UTC, strongest at 04–08 UTC). Splitting on median 5-minute realized volatility, IC remains above 0.34 in both regimes for all symbols, and is slightly stronger in low-volatility conditions — imbalance is more informative when the book is not being disrupted.

Table 5: Per-day rolling IC of `imbalance_qty_11` (mean $\pm$ std across days).

Horizon	BTC	ETH	SOL
1s	+0.442 $\pm$ 0.069	+0.418 $\pm$ 0.099	+0.415 $\pm$ 0.052
5s	+0.433 $\pm$ 0.061	+0.400 $\pm$ 0.098	+0.355 $\pm$ 0.071
30s	+0.260 $\pm$ 0.068	+0.224 $\pm$ 0.072	+0.182 $\pm$ 0.048

4.2 Bootstrap confidence and temporal out-of-sample

A 1000-resample bootstrap on the May 19–21 data gives extremely tight 95% confidence intervals of width ~ 0.02 : for `imbalance_qty_11` at 5 s, BTC [0.442, 0.465], ETH [0.425, 0.450], SOL [0.400, 0.425]. There is no plausible sense in which the signal is noise.

The one cautionary check is temporal out-of-sample (Table 6). Comparing the early window (May 19–21) to the late window (Jun 2–4), the 5 s-horizon IC drops ~ 0.15 on every symbol, while

the 1 s-horizon IC is stable or improves (BTC $0.447 \rightarrow 0.502$). June dates carry wider spreads (0.2–0.3 bps vs. 0.1 bps in May), consistent with thinner books and faster price discovery: the signal is not disappearing, its half-life is shortening.

Table 6: Temporal out-of-sample IC at 5 s (early $\rightarrow$ late).

Feature (5s)	BTC early	BTC late	ETH early	ETH late	SOL early	SOL late
<code>imbalance_qty_l1</code>	+0.453	+0.286	+0.438	+0.277	+0.412	+0.240
<code>imbalance_depth_weighted</code>	+0.435	+0.317	+0.395	+0.270	+0.377	+0.242
<code>raw_ask_depth_5</code>	−0.425	−0.282	−0.395	−0.261	−0.367	−0.233

4.3 Conditional information coefficient: the adverse-selection barrier

The central negative result conditions the IC on whether a directionally-correct mid-cross limit fill would occur within 50 ticks (Table 7).

Table 7: Conditional IC of `imbalance_qty_l1` (BTC/ETH/SOL).

Condition	BTC	ETH	SOL
Unconditional	+0.453	+0.438	+0.412
Any fill event	+0.526	+0.516	+0.460
Buy fill ($\text{imb} > 0$)	+0.032	−0.061	−0.032
Sell fill ($\text{imb} < 0$)	−0.047	−0.027	−0.021

Unconditional IC is 0.41–0.45. Conditioning on the *any*-fill event raises it (0.46–0.53), because ticks near fills carry larger absolute imbalance and hence larger returns. But conditioning on the *directionally-correct* fill — a buy order that actually fills when imbalance predicts up — collapses IC to ~ 0.03 on every symbol. The mechanism is structural:

1. The signal says “buy” ($\text{imbalance} > 0$); a passive buy is posted at the best bid.
2. The fill requires the mid to fall to the bid — an adverse move.
3. At the instant of fill, the “up” prediction has already been invalidated; the post-fill edge is zero.

Fill prevalence rises with book thinness: buy/sell fills occur on 20.8%/21.7% of BTC ticks, 24.1%/24.6% of ETH ticks, and 30.8%/31.1% of SOL ticks. This explains why a naive limit-order simulator produced *more* SOL trades but *worse* PnL — more fills means more adverse selection.

4.4 Validation matrix

Table 8 aggregates the checks against fixed thresholds. The signal passes 7/8 for BTC and 6/8 for ETH/SOL; the failures are the temporal-OOS decay and (for ETH/SOL) the worst-single-day floor.

Verdict. The signal is real, stable, and universal, but non-monetizable under a mid-cross passive fill model. This motivates both an execution-model research track (§8) and an architecture that trades only where the expected move exceeds cost (§5).

Table 8: Validation matrix. Thresholds are fixed *a priori*.

Check	BTC	ETH	SOL	Threshold
Per-day IC mean (5s)	0.43 P	0.40 P	0.35 P	> 0.30
Per-day IC std (5s)	0.06 P	0.10 P	0.07 P	< 0.10
Worst single day (5s)	0.24 P	0.04 F	0.17 F	> 0.20
Intraday (all windows)	P	P	P	all > 0.20
Low-vol IC	0.46 P	0.43 P	0.39 P	> 0.30
High-vol IC	0.42 P	0.39 P	0.34 P	> 0.30
Bootstrap CI lower	0.44 P	0.43 P	0.40 P	> 0.30
Temporal OOS delta	0.17 F	0.16 F	0.17 F	< 0.10
Total	7/8	6/8	6/8	

5 Hierarchical Signal Combination

5.1 Architecture

The hierarchical combiner separates eleven features from the scan into three functional layers and composes them multiplicatively:

Layer 1 — slow directional bias Inputs `regime_divergence_1h`, `spread_bps`, `trend_ema_short`. IC-weighted rolling z-score, thresholded to $\{-1, 0, +1\}$; active only when $|z| > 0.5$.

Layer 2 — fast entry timing Inputs `OBI_11`, `cross_obi`, `queue_position_bid`, `vwap_deviation`, `obi_velocity`. IC-weighted rolling z-score clipped to $[-1, +1]$, *zeroed when its sign disagrees with Layer 1*.

Layer 3 — volatility sizing Inputs `hawkes_intensity`, `flow_count_30s`, `vol_returns_5m`. IC-weighted z-score through a sigmoid to $[0, 1]$, scaling position *inversely* with volatility.

The composite is

$$S = L_1^{\text{dir}} \times |L_2^{\text{entry}}| \times L_3^{\text{scale}}, \quad S = 0.3 |L_2^{\text{entry}}| L_3^{\text{scale}} \text{ when } L_1 \text{ neutral.} \quad (2)$$

Rationale. Three design choices follow directly from §4: (i) IC-magnitude weights (rather than a learned model) are more robust with only 3–5 features per layer; (ii) directional gating is the key mechanism — fast signals alone are non-monetizable (conditional IC ~ 0.03), but fast signals *confirming* a slow bias operate at horizons where the expected move exceeds execution cost; (iii) inverse-volatility sizing reduces exposure precisely when adverse selection on passive orders is worst.

5.2 Walk-forward out-of-sample results

Training uses a 4-fold expanding window with a 100-bar embargo on 5-minute bars. Table 9 reports per-fold and mean composite IC, directional accuracy, and cost-adjusted Sharpe (BTC/ETH at a 60-bar ≈ 5 h horizon; SOL at 40 bars ≈ 3.3 h due to less data).

Layer-1 is active 42–45% of the time. The learned IC weights closely track the scan priors (e.g. BTC L1: `regime_divergence` +0.071, `spread_bps` +0.138, `trend_ema` −0.142), which argues against pure walk-forward overfitting.

5.3 Comparison and honest assessment

Against the 30-day `nat oos30` benchmark at the 100-minute horizon, the combiner (Sharpe 1.25/1.71/2.40) is weaker than the deployable winners — `3f_liquidity` (9.2/7.8/5.1), `jump_detector` (4.1/3.8/6.2),

Table 9: Hierarchical combiner walk-forward OOS results.

Symbol	Metric	Fold 0	Fold 1	Fold 2	Fold 3	Mean
BTC	Composite IC	+0.077	+0.179	+0.224	+0.253	+0.178
	Dir. accuracy	0.543	0.552	0.563	0.571	0.557
	Sharpe (cost-adj.)	+0.41	+1.12	+1.58	+1.87	+1.25
ETH	Composite IC	+0.156	+0.210	+0.278	+0.349	+0.248
	Dir. accuracy	0.558	0.571	0.582	0.591	0.576
	Sharpe (cost-adj.)	+0.92	+1.38	+2.01	+2.54	+1.71
SOL	Composite IC	+0.212	+0.318	+0.401	+0.504	+0.359
	Dir. accuracy	0.572	0.588	0.601	0.615	0.594
	Sharpe (cost-adj.)	+1.35	+2.08	+2.71	+3.45	+2.40

optimal_entry (3.5/2.9/4.7), **funding_reversion** (2.8/3.1/1.9). The comparison is unfair in both directions: the combiner used only two days of data (vs. thirty) and a different horizon, but it is the only architecture that explicitly addresses adverse selection.

Promising. Consistent positive OOS IC on all three symbols with cost-adjusted Sharpe > 0 ; a layer structure matching the orthogonality of §3; working directional gating (Layer-2 IC conditional on Layer-1 alignment exceeds its unconditional IC); weight stability versus the scan priors.

Concerning. Only two days of data, so every figure carries wide intervals. The monotone fold progression ($\text{fold}_0 < \text{fold}_1 < \text{fold}_2 < \text{fold}_3$) is suspicious and may indicate look-ahead or a trending window. Layer-1 dominates the composite, so the “hierarchical” benefit over Layer-1 alone is not yet demonstrated. The SOL figures (shorter horizon, less data) are an upper bound. The 11 bps round-trip cost is a configuration assumption, not a measured fill. And at a 5h effective horizon this is arguably a medium-frequency momentum signal rather than a microstructure one. The required next steps are data accumulation, an L1-only ablation, conditional-fill analysis on real logs, and a horizon cross-validation.

6 Complementary Pattern Discovery: The Convolver

A separate engine discovers *candle-shape* patterns via matched filters: events (breakouts, turtle-soup reversals, traps) are detected, stacked into channel matrices, and reduced by truncated SVD (Eckart–Young low-rank approximation) into kernels; kernels are admitted by an IC gate and a Benjamini–Hochberg false-discovery-rate correction. The methodology is sound; the constraint is data volume.

6.1 Current state and sample-size arithmetic

BTC discovery (2026-05-24) used 12,059 60-second candles (≈ 8.4 days) and found 1,775 events across six types, of which 6 kernels survived FDR but 0/6 were out-of-sample robust (walk-forward decay ratio < 0.50). Temporal stability of the surviving shapes was nonetheless high (turtle-bull $\rho = 0.920$, trap-bull $\rho = 0.941$). Table 10 gives the event rates; trap events are the bottleneck at 11–14 per 1000 candles.

SVD stability needs ≥ 100 events per (event-type, channel) pair; with four channels, trap-bull’s 136 events give only ~ 34 per channel, below threshold. Reliable walk-forward IC needs ≥ 200 OOS events, i.e. ~ 500 total per type — about 39,000 candles for traps. The “50K candle” recommendation yields ~ 600 trap events (~ 240 OOS): marginal but workable on *sample size*.

Table 10: Observed BTC event rates (12,059 candles, 60s).

Event type	Count	Per 1000 candles	Per day
breakout_bull	418	34.7	~50
breakout_bear	522	43.3	~62
turtle_bull	286	23.7	~34
turtle_bear	241	20.0	~29
trap_bull	136	11.3	~16
trap_bear	172	14.3	~21
Total	1,775	147.2	~212

6.2 Why 50K candles is necessary but not sufficient

Sample size is only one constraint. **Regime coverage** is the deeper one: 50K 60s-candles span ~ 35 days, capturing only 1–2 market regimes, so the walk-forward test is within-regime rather than across-regime. A pattern discovered during low-volatility accumulation may simply *not occur* during a liquidation cascade. To prove regime-robustness the OOS window must straddle a regime transition, requiring ~ 6 months minimum. **Multi-timeframe scaling** compounds this: event rates fall linearly with candle size (Table 11). **Non-stationarity** of event rates and **pattern decay** (market-maker, fee, and participant changes over months/years) create an irreducible tension between “more data” and “stale data”; the proposed mitigation is exponential event weighting with a 3–6 month half-life, not yet implemented.

Table 11: Calendar time to collect 200 OOS trap events by candle size.

Candle	Candles/day	Trap rate/day	Calendar time
60s	1,440	~17	~5 weeks
5min	288	~3.5	~5 months
15min	96	~1.2	~1.5 years
1hr	24	~0.3	~5 years
4hr	6	~0.07	~20 years

6.3 Viable niche and use as ML features

The convolver’s defensible niche is short-timescale (60s, 20-minute window) patterns on high-frequency data — complementary to tick-level signals because it operates on candle *shape*, not order-book state. Crucially, its eight `alg_conv_*` outputs are expected to be decorrelated from all microstructure features ($|\rho| < 0.2$) because they derive from a different data representation. This makes them high-value *inputs* to the machine-learning algorithms rather than a standalone strategy — the convolver need not pass the trading gauntlet independently if its features improve the models that consume them. Finally, a 0-trade gauntlet result on June 1 was traced to a configuration bug (bar aggregation `mean` dilutes a score that is 0 between candles and spikes briefly; the fix is `max` aggregation), not a signal failure.

7 System Context

These findings sit inside a mature platform (~ 120 K LOC: a 4-crate Rust ingestor of ~ 43 K LOC computing 236 features at $p99 < 80$ ms/tick, and ~ 70 K LOC of Python). The feature engine, ingestor, GMM regime classifier, five-hypothesis test suite (all of H1–H5 confirmed), a 34+ algorithm library (five deployable winners), four autonomous research agents with Benjamini–Hochberg

FDR control, a nine-stage alpha pipeline, and an eleven-service Docker stack are all production-grade. Against this backdrop the assessment is unambiguous: *the binding constraint is no longer signal quality but execution feasibility*. A validated $IC = 0.45$ directional signal exists; it drops to ~ 0.03 conditional on mid-cross fills. Raw per-trade persistence (price/size/side/timestamp) was landed on 2026-06-09 specifically to enable an event-driven fill model that replaces the mid-cross approximation.

8 Discussion

The results invert the usual order of difficulty. Discovering a strong, robust, universal directional signal turned out to be the easy part; the entire economic question reduces to whether that signal can be captured net of adverse selection. Three implications follow.

Execution is the frontier. The conditional-IC result (§4.3) is a property of the *fill model*, not the signal. The defined next steps are: (i) build a `load_trades()` utility over the new trade Parquet; (ii) construct an event-driven fill model from actual aggressor trades; (iii) re-measure conditional IC under that model — if it exceeds ~ 0.15 , the signal is tradeable; (iv) only then combine the eight axes and paper-trade.

The signal is shortening. The ~ 0.15 five-second decay from May to June, with a stable one-second IC, indicates that the tradeable horizon is contracting as books thin. Any deployment must track this and adapt stale-signal timeouts per symbol (SOL fastest at a ~ 15 s half-life).

Two complementary tracks. The hierarchical combiner and the convolver attack different timescales: the combiner gates microstructure timing on a slow bias to clear execution cost at multi-minute-to-hour horizons; the convolver supplies decorrelated candle-shape features to the ML layer. Neither is yet deployable — the combiner needs more data and an ablation; the convolver needs 6–24 months and recency weighting — but both are structurally aligned with the orthogonality the data exhibits.

9 Conclusion

Order-book imbalance predicts the direction of Hyperliquid perpetual returns with $IC \approx 0.45$ at 1–5 seconds, universally across BTC/ETH/SOL, surviving an eight-check robustness battery. The same data show, decisively, that a passive mid-cross fill model eliminates the edge through adverse selection ($IC \rightarrow 0.03$). A hierarchical combiner that gates fast timing on a slow directional bias produces positive, cost-adjusted out-of-sample performance on a thin window and is the first architecture here to confront adverse selection directly. A matched-filter pattern-discovery engine is sound but data-starved. The research programme’s binding constraint is therefore execution feasibility, and the immediate path is an event-driven fill model built on newly-collected per-trade data.

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